

```
manchodu@Srini:~/OSLAB$ nano Signal1.c
manchodu@Srini:~/OSLAB$ gcc signal1.c -o signal1
cc1: fatal error: signal1.c: No such file or directory
compilation terminated.
manchodu@Srini:~/OSLAB$ nano Signal1.c
manchodu@Srini:~/OSLAB$ gcc Signal1.c -o Signal1
manchodu@Srini:~/OSLAB$ ./Signal1
Program is running...

Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
^C
SIGINT received!
Program is running...
Program is running...
^C
SIGINT received!
Program is running...
^C
SIGINT received!
Program is running...
Program is running...
Program is running...
|
```

```
manchodu@Srini:~$ cd OSLAB
manchodu@Srini:~/OSLAB$ nano Signal2.c
manchodu@Srini:~/OSLAB$ gcc Signal2.c -o Signal2
manchodu@Srini:~/OSLAB$ ./Signal2
Program is running...
Program is running...
Program is running...
^C
```

```
manchodu@Srini:~/OSLAB$ nano Signal3.c
manchodu@Srini:~/OSLAB$ gcc Signal3.c -o Signal3
manchodu@Srini:~/OSLAB$ ./Signal3
Parent is waiting...
Child is running...
^C
manchodu@Srini:~/OSLAB$ |
```

```
manchodu@Srini:~/OSLAB$ nano Signal4.c
manchodu@Srini:~/OSLAB$ gcc Signal4.c -o Signal4
manchodu@Srini:~/OSLAB$ ./Signal4
Writing to pipe...
manchodu@Srini:~/OSLAB$ |
```

```
manchodu@Srini:~/OSLAB$ nano Signal5.c
manchodu@Srini:~/OSLAB$ gcc Signal5.c -o Signal5
manchodu@Srini:~/OSLAB$ ./Signal5
Child waiting...
Parent sending signal...
Child received SIGUSR1
manchodu@Srini:~/OSLAB$ |
```

```
manchodu@Srini:~/OSLAB$ nano Signal6.c
manchodu@Srini:~/OSLAB$ gcc Signal6.c -o Signal6
manchodu@Srini:~/OSLAB$ ./Signal6
Program started
Aborted (core dumped)
manchodu@Srini:~/OSLAB$ |
```

```
manchodu@Srini:~/OSLAB$ nano Signal7.c
manchodu@Srini:~/OSLAB$ gcc Signal7.c -o Signal7
manchodu@Srini:~/OSLAB$ ./Signal7
Before error
Segmentation fault (core dumped)
manchodu@Srini:~/OSLAB$ |
```

```
manchodu@Srini:~/OSLAB$ cat Signal1.c
#include <stdio.h>
#include <signal.h>
#include <unistd.h>
void handle_signal(int sig)
{
printf("\nSIGINT received!\n");
}
int main()
{
signal(SIGINT, handle_signal);
while (1)
{
printf("Program is running...\n");
sleep(2);
}
return 0;
}
```

```
manchodu@Srini:~/OSLAB$ cat Signal2.c
#include <stdio.h>
#include <unistd.h>
int main()
{
while (1)
{
printf("Program is running...\n");
sleep(2);
}
return 0;
}
manchodu@Srini:~/OSLAB$
```

```
manchodu@Srini:~/OSLAB$ cat Signal3.c
#include <stdio.h>
#include <signal.h>
#include <unistd.h>
#include <sys/wait.h>
void handle_sigchld(int sig)
{
printf("SIGCHLD received: Child has terminated.\n");
}
int main()
{
signal(SIGCHLD, handle_sigchld);
pid_t pid = fork();
if (pid == 0)
{
// Child
printf("Child is running...\n");
sleep(3);
printf("Child is terminating...\n");
}
else
{
// Parent
printf("Parent is waiting...\n");
wait(NULL);
printf("Parent finished.\n");
}
return 0;
}
manchodu@Srini:~/OSLAB$
```

```
manchodu@Srini:~/OSLAB$ cat Signal4.c
#include <stdio.h>
#include <unistd.h>
int main()
{
    int fd[2];
    pipe(fd);
    close(fd[0]); // Close read end
    printf("Writing to pipe...\n");
    write(fd[1], "Hello", 5);
    printf("Write completed.\n");
    return 0;
}
```

```
manchodu@Srini:~/OSLAB$ cat Signal5.c
#include <stdio.h>
#include <signal.h>
#include <unistd.h>
#include <sys/wait.h>
void handler(int sig)
{
    printf("Child received SIGUSR1\n");
}
int main()
{
    pid_t pid = fork();
    if (pid == 0)
    {
        // Child
        signal(SIGUSR1, handler);
        printf("Child waiting...\n");
        while (1)
        {
            sleep(1);
        }
    }
    else
    {
        // Parent
        sleep(2);
        printf("Parent sending signal...\n");
        kill(pid, SIGUSR1);
        sleep(2);
        kill(pid, SIGTERM);
        wait(NULL);
    }
    return 0;
}
```

```
manchodu@Srini:~/OSLAB$ cat Signal6.c
#include <stdio.h>
#include <stdlib.h>
int main()
{
printf("Program started\n");
abort();
printf("Program ended\n");
return 0;
}
```

```
manchodu@Srini:~/OSLAB$ cat Signal7.c
#include <stdio.h>
int main()
{
int *p = NULL;
printf("Before error\n");
*p = 10;
printf("After error\n");
return 0;
}
```